EXAM, LINEAR ANALYSIS, MARCH 16 2022.

(Allowed tools: Pen or pencil, rubber.) The exam time is 5 hours. No cheating.

Remember to explain all non-trivial identities as clearly as possible.

Problem 1

Find the radius of convergence of the following power series:

a) $\sum_{n=1}^{\infty} n^4 x^n$
b) $\sum_{n=5}^{\infty} (2-1/n)^n x^n$
c) $\sum_{n=1}^{\infty} e^{n^2} x^n$

Solution: a) We have

$$\lim_{n\to\infty} \frac{(n+1)^4}{n^4} = \lim_{n\to\infty} \left(\frac{n+1}{n}\right)^4 = \left(\lim_{n\to\infty} \frac{n+1}{n}\right)^4,$$

since $x \mapsto x^4$ is a continuous function. Thus the limit equals 1 and this is also the radius of convergence, by the quotient test.

b) We have

$$\lim_{n\to\infty} \left((2-1/n)^n\right)^{1/n} = 2,$$

so the radius of convergence is 1/2 by the root-test (a.k.a. Cauchy-Hadamard test).

c) For any $x > 0$ we have

$$\lim_{n\to\infty} e^{n^2} x^n = \lim_{n\to\infty} e^{n^2 + n\ln x} = e^{\lim_{n\to\infty} n^2 + n\ln x} = \infty,$$

so the radius of convergence equals 0. (We could also have used the root-test as above...)

Problem 2

Find a solution to the following problem:

$$\begin{cases} \partial_t u(x,t) = 2\partial_{xx}^2 u(x,t), & 0 < x < \pi,\ t > 0 \\ u(0,t) = u(\pi,t) = 0, & t > 0 \\ u(x,0) = 1, & 0 < x < \pi \end{cases}$$

Solution: We first guess a separated solution $u(x,t) = X(x)T(t)$, and conclude that $\frac{T'(t)}{T(t)} = 2\frac{X''(x)}{X(x)}$ is constant. Takin the boundary conditions into account this leads to solutions $X(x) = \sin(nx)$ for any $n \in \mathbb{N}$. Corresponding solutions for the T-equation gives $T(t) = e^{-2nt}$ so we find that

$$u_n(x,t) = \sin(nx)e^{-2nt}$$

are solutions to the first two equations. Since we know that any "decent" function, in particular the function $f(x) = 1$ on $[0,\pi]$, can be expressed as a sine-series (despite the fact that $f(0) \neq 0$!), we claim that a solution to the full problem is

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin(nx) e^{-2nt}$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} \sin(nx) = \begin{cases} \frac{4}{\pi n} & n\text{odd} \\ 0 & n\text{even} \end{cases}$$

To see that this really solves the problem, we note that f is bounded and continuous at each x in $(0,\pi)$, so by the general theory of Fourier series the series converges to 1 at each $(x,0)$. Thus $u(x,0) = 1$ holds. For $t > 0$ we can consider e^{-2nt} as part of the coefficient in the sine-series and conclude that the series is absolutely convergent, hence $u(0,t) = u(\pi,t) = 0$. Finally, to prove that u really solves the partial differential equation, we need to verify that

$$\partial_t \sum_{n=1}^{\infty} b_n \sin(nx) e^{-2nt} = \sum_{n=1}^{\infty} \partial_t b_n \sin(nx) e^{-2nt} = \sum_{n=1}^{\infty} -2nb_n \sin(nx) e^{-2nt},$$

as well as a similar formula for the ∂^2_{xx}-derivative. By the theory of uniformly convergent series, this is true (for any given fixed x) as long as the right hand side is uniformly convergent. Fix $\epsilon > 0$ and note that for $t > \epsilon$ we have

$$\sum_{n=1}^{\infty} |-2nb_n \sin(nx) e^{-2nt}| \leqslant \sum_{n=1}^{\infty} \frac{8}{\pi} e^{-\epsilon n} = \frac{8}{\pi}\frac{1}{1-e^{-1}} < \infty$$

so the uniform convergence follows by the Weierstrass M-test. The analogous argument for ∂^2_{xx} is similar so we omit the details.

Problem 3

Let f be the 2π-periodic function given by $f(x) = x^2$ for $-\pi < x \leqslant \pi$.

a) Find the Fourier series for f.
b) Does the series converge uniformly to f?
c) Find the value of the series

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}.$$

(*Hint: You may need to use that* $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.)

d) Find the value of the series

$$\sum_{n=1}^{\infty} \frac{1}{n^4}.$$

Solution: For $n \neq 0$ we get

$$\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 e^{-inx} dx = \ldots \text{(partial integration)} \ldots = 2\frac{(-1)^n}{n^2}$$

whereas $\hat{f}(0) = \frac{\pi^2}{3}$. Since f is continuous at all points (!), we have

$$f(x) = \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{inx} = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2}\cos(nx),$$

by the Fourier Inversion Theorem. The convergence is uniform since $\sum_{n=-\infty}^{\infty} |\hat{f}(n)| < \infty$. Evaluating at $x = 0$ gives

$$0 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{(2n)^2} - \sum_{n=0}^{\infty} \frac{4}{(2n+1)^2} = \frac{\pi^2}{3} + \frac{\pi^2}{6} - \sum_{n=0}^{\infty} \frac{4}{(2n+1)^2}$$

which leads to $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$. Finally, by Parseval's identity we get

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} x^4 dx = \sum_{n=-\infty}^{\infty} |\hat{f}(n)|$$

which gives $\frac{\pi^4}{5} = \frac{\pi^4}{9} + 2\sum_{n=1}^{\infty} \frac{4}{n^4}$ and hence $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$.

Problem 4

Find the polynomial $p(x)$ of degree $\leqslant 2$ that minimizes

$$\int_0^1 |e^x - p(x)|^2 \, dx.$$

Solution: We first need to compute an orthonormal basis for the linear subspace of polynomials of degree $\leqslant 2$, using Gram-Schmidt on $\{1, x, x^2\}$. After some quite tedious computations we arrive at

$$p_0 = 1, \quad p_1 = 2\sqrt{3}(x - 1/2), \quad p_2 = 6\sqrt{5}(x^2 - x + 1/6).$$

By the "least distance lemma" the closest polynomial in the subspace is

$$P(e^x) = \langle e^x, p_2 \rangle p_2 + \langle e^x, p_1 \rangle p_1 + \langle e^x, p_0 \rangle p_0.$$

Some more computations yield $\langle e^x, p_0 \rangle = e - 1$, $\langle e^x, p_1 \rangle = \sqrt{3}(3 - e)$ and $\langle e^x, p_2 \rangle = \sqrt{5}(7e - 19)$ so the sought polynomial is

$$(210e - 570)x^2 - (216e - 588)x + (39e - 105).$$

See figure 1 for an illustration.

Problem 5

Let f be the function given by

$$f(x) = \sum_{n=1}^{\infty} e^{-nx}, \quad x > 0.$$

Prove that

a) the series converges uniformly on each compact subinterval of $(0, \infty)$.
b) f is differentiable at each $x > 0$ and compute $f'(x)$.
c) $\int_0^{\infty} f(x)dx = \infty$.

Solution: a) Let $\epsilon > 0$ be arbitrary but fixed. For $x \geqslant \epsilon$ we have that $e^{-nx} \leqslant e^{-n\epsilon}$ and $\sum_{n=1}^{\infty} e^{-n\epsilon}$ converges so the convergence is uniform on $[\epsilon, \infty)$ by the Weierstrass M-test.

b) and c) are most easily solved by noting that

$$\sum_{n=1}^{\infty} e^{-nx} = \sum_{n=0}^{\infty} (e^{-x})^n - 1 = \frac{e^{-x}}{1 - e^{-x}},$$

and then no further theory is needed. If one fails to recognize this (indeed, so did the maker of the exam before it was too late), we solve *b*) by noting that the formally differentiated series is $\sum_{n=1}^{\infty} -ne^{-nx}$, which also converges uniformly on compacts (by a similar argument as in *a*)). Hence by the theory of uniformly convergent series we have $f'(x) = -\sum_{n=1}^{\infty} ne^{-nx}$. For *c*) we can not immediately use the theory to conclude that

$$\int_0^{\infty} f(x)dx = \sum_{n=1}^{\infty} \int_0^{\infty} e^{-nx} dx = \sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

because we do not have uniform convergence on $(0, \infty)$ (the above identity is true, but it takes a bit more to prove it, most easily it follows from the monotone convergence theorem in the integration

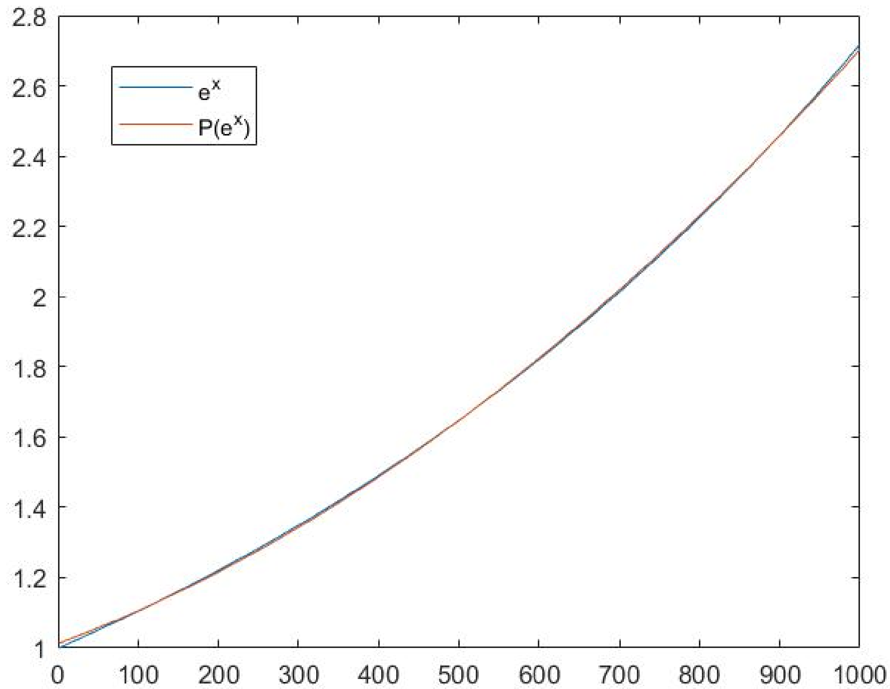

FIGURE 1. Plot of e^x and its best approximation by a polynomial of degree 2 on the interval $[0,1]$ (Nevermind the x-axis label in the figure...).

theory course). To get around this obstacle we can realize that $f(x) \geqslant \sum_{n=1}^{N} e^{-nx}$ for every N and thus we get

$$\int_0^\infty f(x)dx \geqslant \sum_{n=1}^{N} \int_0^\infty e^{-nx} dx = \sum_{n=1}^{N} \frac{1}{n},$$

which gives that the integral must be infinite since the right hand side can be made arbitrarily large.

PROBLEM 6

Let a_n be the Fourier coefficients of the function in Problem 3, and let b_n be the Fourier coefficients of the $2\pi-$periodic function which equals x^2 on $[0, 2\pi]$. Prove, without computing the b_n's,

a) that $(\frac{b_n}{a_n})_{n=1}^{\infty}$ is an unbounded sequence.
   (*Hint: You may use that* $a_n = O(1/n^2)$ *as* $n \to \pm\infty$.)
b) that $\sum_{n=-\infty}^{\infty} b_n = 2\pi^2$.

Also explain why it is natural to expect that the a_n's go faster to 0 than the b_n's, as $n \to \pm\infty$.

Solution: *a)* Let $g(x)$ denote the function under consideration. If $(\frac{b_n}{a_n})_{n=1}^{\infty}$ had been bounded we would have had $|b_n| = O(1/n^2)$ as $n \to \pm\infty$. In this case the Fourier series would be uniformly convergent (since $\sum_{n=1}^{\infty} 1/n^2 < \infty$) and then the function g would be continuous (since $g(x) = \sum_{n=-\infty}^{\infty} b_n e^{inx}$ at each point of continuity of g, by the Fourier inversion theorem). However, g clearly has a jump discontinuity at $x = 0$.

b) At $x = 0$ the function has a jump discontinuity of size $4\pi^2$. Also the left and right derivatives exist. By the Fourier Inversion Theorem the Fourier series will converge to the mean of the left and right function value, i.e. $(4\pi^2 - 0)/2 = 2\pi^2$.

Finally, we should anticipate that the a_n's decay faster than the b_n's because the function f in Problem 3 is continuous whereas g is not, and we know that fast decay of the coefficients goes hand in hand with smoothness properties of the corresponding function.